A Unified Benchmarking Pipeline for Satellite Image Restoration

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***Abstract*—Deep learning models are widely used for satellite image restoration tasks, but their evaluation varies widely. This makes results difficult to reproduce and complicate the integration of new datasets, models, and evaluation outputs. This paper presents a modular benchmarking pipeline for satellite image restoration that brings dataset loading, preprocessing, model execution, metric computation, visualization, and experiment logging into a single structured workflow. The current framework is evaluated on two representative restoration settings: cloud removal using AllClear and super-resolution using PROBA-V SR. Our implementation reports reconstruction-oriented metrics including Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) along others, and reports experiment configurations together with quantitative and visual outputs. The results demonstrate the practical use of the framework as an evaluation tool, while also identifying the need for broader benchmark coverage and additional baselines in future work. The project source code can be accessed via the GitHub repository at https://github.com/Royana-Huseynova/SDP-2026**

***Index Terms*—satellite image restoration, benchmarking pipeline, cloud removal, super-resolution, remote sensing, deep learning, reproducibility, evaluation metrics, experiment logging**

I. Introduction

Deep learning methods for satellite image restoration have advanced rapidly across tasks such as cloud removal and super-resolution. However, their evaluation remains fragmented, as existing benchmarks are typically task-specific and rely on different datasets, preprocessing pipelines, and metric implementations [2], [4]. This makes results harder to reproduce and can make model comparison less transparent.

To address this problem, this paper presents a modular, reproducible, and scalable benchmarking pipeline for satellite image restoration. The framework integrates dataset loading, preprocessing, model execution, metric computation, visualization, and experiment logging into a single evaluation workflow. In its current implementation, it supports AllClear for cloud removal and PROBA-V SR for super-resolution, while preserving the task-specific requirements of each dataset.

The proposed framework is intended to support clearer model comparison, easier integration of new methods, and more transparent experimental reporting. By providing a shared evaluation structure, it helps reduce methodological inconsistency across restoration tasks and offers a practical foundation for future benchmarking research in remote sensing. The main contributions of this paper are as follows:

- We design a modular benchmarking pipeline for satellite image restoration that integrates dataset loading, preprocessing, model execution, metric computation, visualization, and experiment logging into a single evaluation workflow.
- We demonstrate the pipeline on two representative restoration settings: cloud removal using AllClear and super-resolution using PROBA-V SR. The framework preserves task-specific dataset and model requirements while using a shared structure for evaluation and output generation.
- We provide a reproducible experiment structure that stores quantitative metrics, visual outputs, and configuration records together, making restoration experiments easier to inspect, repeat, and extend.

The rest of this paper is organized as follows. Section II reviews related work on satellite image restoration, including super-resolution, cloud removal, and corresponding evaluation frameworks. Section III presents the methodological foundation of the proposed system, including the datasets, task definitions, evaluation metrics, and reproducibility and scalability considerations. Section IV describes the experimental setup used to assess the framework. Section V presents the quantitative and qualitative results and discusses their implications. Finally, Section VI concludes the paper and outlines directions for future work.

II. Related Work

Research on satellite image restoration includes task-specific problems such as super-resolution and cloud removal, as well as evaluation studies that examine how restoration quality should be measured. Super-resolution and cloud removal are both restoration problems, but they differ in objective, input structure, and evaluation assumptions. For this reason, the present work does not treat them as identical tasks. Instead, it uses them as two representative settings for demonstrating a shared benchmarking pipeline.

A. Satellite Image Super-Resolution

Satellite image super-resolution aims to reconstruct higher-resolution outputs from lower-resolution observations. The PROBA-V Super-Resolution Challenge is a widely used benchmark for this task. It asks participants to fuse multiple low-resolution images of the same Earth location into a single super-resolved image, which is then compared with a higher-resolution reference image. Building on this setting, Nguyen et al. proposed Proba-V-ref, which extends PROBA-V super-resolution through reference-aware reconstruction from related scene observations [1]. These works helped define a clear benchmark setting for satellite super-resolution, while also showing the practical difficulties of the task, including image registration, temporal variation, cloud contamination, and reference quality.

Recent satellite super-resolution methods include convolutional neural network (CNN), generative adversarial network (GAN), and Transformer-based approaches [2]. Surveys of remote sensing super-resolution show that evaluation is sensitive to dataset choice, preprocessing, metric selection, and the availability of reliable reference images [2], [3]. These issues make standardized evaluation important even within a single restoration task.

B. Cloud Removal Methods

Cloud removal is another important restoration problem in optical satellite imaging because clouds and cloud shadows can hide land-surface information. Recent cloud-removal methods often use multi-temporal or multi-modal information to recover cloud-covered regions. Zhou et al. introduced AllClear, a large-scale benchmark for satellite cloud removal that combines optical imagery, synthetic aperture radar (SAR), cloud masks, and other remote sensing products [4]. Xu et al. proposed the M3R-CR dataset and AlignCR model for cloud removal through progressive alignment of SAR and optical features [5].

Other works further show the diversity of cloud-removal settings. McGANs use multispectral imagery and conditional adversarial learning for thin-cloud removal [6], while SEN12MS-CR-TS provides a multi-modal and multi-temporal dataset for cloud removal using Sentinel-1 and Sentinel-2 observations [7]. These studies demonstrate strong progress in cloud removal, but they also show that evaluation settings can differ substantially in input modality, temporal structure, cloud severity, and target definition.

C. Evaluation Metrics and Benchmarking Practice

Evaluation in satellite image restoration commonly uses both general image-quality metrics and remote sensing-oriented metrics. Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) are widely used to measure pixel-level fidelity and structural similarity. Spectral Angle Mapper (SAM) and Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) are commonly used in remote sensing to assess spectral consistency and relative reconstruction error [8]. Learned Perceptual Image Patch Similarity (LPIPS) is also useful when restoration quality should be evaluated in terms of perceptual similarity rather than only pixel-wise agreement. These metrics are useful because they measure different aspects of restoration quality, but no single metric fully describes the usefulness of a restored satellite image.

Task-aware evaluation has therefore become important. For example, Liu et al. introduced SAGE-NDVI to evaluate whether restored imagery preserves agriculturally meaningful vegetation information [8]. This reflects a broader issue in restoration benchmarking: high numerical similarity does not always guarantee visual realism or downstream usefulness. Evaluation also becomes harder when reference images are imperfect, temporally mismatched, or affected by different preprocessing assumptions [9], [10].

III. Methodology

This paper presents a benchmarking pipeline rather than a new restoration model. The goal is to organize the main stages of satellite image restoration evaluation into a single reproducible workflow. The pipeline includes dataset loading, preprocessing, model execution, metric computation, visualization, and experiment logging. It is designed to support different restoration tasks without forcing them into the same data format or model interface. In the present work, cloud removal and super-resolution are treated as two representative satellite restoration settings that share common evaluation needs. In this setting, AllClear [1] dataset is used for cloud removal, while PROBA-V SR [4] is used for satellite image super-resolution. The framework preserves task-specific dataset and model requirements, but standardizes the surrounding evaluation workflow.

A. Framework Overview

The framework follows a sequential evaluation process. First, a task-specific dataset loader reads the selected benchmark data. Second, preprocessing prepares the input data for the selected model. Third, the model is executed through a shared inference interface. Fourth, the prediction is compared with the reference target using predefined metrics. Finally, the framework stores metric values, visual outputs, and configuration records.

This design separates shared evaluation components from task-specific components. Shared components include metric computation, logging, visualization, and output organization. Task-specific components include dataset loading, input preparation, model adapters, and restoration targets. This separation allows the framework to remain reusable while still respecting the differences between cloud removal and super-resolution.

B. Supported Tasks and Datasets

The current implementation supports two restoration tasks. In cloud removal, the goal is to reconstruct image content affected by clouds or cloud shadows. The framework uses AllClear for this setting. Depending on the model, the input may include cloudy optical images, temporal observations, cloud masks, or auxiliary modalities.

In super-resolution, the goal is to recover a higher-resolution image from lower-resolution observations. The framework uses PROBA-V SR for this setting. The dataset requires different loading and preprocessing logic from cloud removal because the task is based on multi-image super-resolution rather than cloud-contaminated image reconstruction.

The two datasets are therefore not converted into one identical format. Instead, each dataset is handled through its own loader and then passed into a common evaluation workflow. This keeps the benchmark definitions intact while allowing the framework to reuse the same evaluation, logging, and visualization structure.

C. Preprocessing and Model Execution

Preprocessing is treated as part of the benchmark rather than as an external step. This is important because satellite datasets may differ in spatial resolution, spectral bands, valid-data masks, temporal structure, and file organization. The framework records preprocessing choices so that experiments can be inspected and repeated.

For each task, preprocessing may include normalization, band selection, band alignment, resizing, mask handling, and conversion into model-compatible tensors. After preprocessing, the selected model is executed through a shared inference interface. The interface passes prepared inputs to the model, collects predictions, and stores outputs for evaluation. Model-specific requirements are handled through adapters, while the overall execution procedure remains consistent.

D. Evaluation Metrics

The framework evaluates predictions using well-known reconstruction-oriented metrics. Let Y denote the reference image and $\hat{Y}$ denote the restored image. For an image with N pixels, Peak Signal-to-Noise Ratio (PSNR) measures pixel-level reconstruction fidelity:

$$\mathrm{PSNR}(Y,\hat{Y}) = 10\log_{10}\left(\frac{L^2}{\mathrm{MSE}(Y,\hat{Y})}\right), \tag{1}$$

where L is the maximum possible pixel value, and MSE is mean squared error. Higher PSNR indicates lower pixel-level error. Often, corrected PSNR (cPSNR) is preferred over standard PSNR because it measures reconstruction quality only in the corrupted or masked region, rather than letting unchanged background pixels artificially inflate the score. Structural Similarity Index Measure (SSIM) compares local luminance, contrast, and structure:

$$\mathrm{SSIM}(Y,\hat{Y}) = \frac{(2\mu_Y\mu_{\hat{Y}} + C_1)(2\sigma_{Y\hat{Y}} + C_2)}{(\mu_Y^2 + \mu_{\hat{Y}}^2 + C_1)(\sigma_Y^2 + \sigma_{\hat{Y}}^2 + C_2)}, \tag{2}$$

where μ_Y and $\mu_{\hat{Y}}$ are local means, σ_Y^2 and $\sigma_{\hat{Y}}^2$ are local variances, $\sigma_{Y\hat{Y}}$ is the local covariance, and C_1, C_2 are stabilizing constants. Higher SSIM indicates greater structural similarity. Spectral Angle Mapper (SAM) measures the spectral angle between the reference and restored spectral vectors at each pixel. For spectral vectors y_i and $\hat{y}_i$, SAM is defined as

$$\mathrm{SAM}(Y,\hat{Y}) = \frac{1}{N}\sum_{i=1}^{N} \arccos\left(\frac{y_i^\top \hat{y}_i}{\|y_i\|_2\|\hat{y}_i\|_2}\right). \tag{3}$$

Lower SAM indicates better spectral consistency. Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) measures relative global reconstruction error across spectral bands:

$$\mathrm{ERGAS}(Y,\hat{Y}) = 100\frac{h}{l}\sqrt{\frac{1}{B}\sum_{b=1}^{B}\left(\frac{\mathrm{RMSE}_b}{\mu_b}\right)^2}, \tag{4}$$

where h/l is the ratio between high-resolution and low-resolution pixel sizes, B is the number of spectral bands, RMSE_b is the root mean squared error for band b, and μ_b is the mean value of the reference image in band b. Lower ERGAS indicates lower relative reconstruction error. **Learned Perceptual Image Patch Similarity (LPIPS)** measures perceptual distance using deep feature activations rather than direct pixel differences. Let $\phi_l(Y)$ and $\phi_l(\hat{Y})$ denote normalized feature maps extracted from layer l of a pretrained network, and let w_l denote the learned channel-wise weights for that layer. LPIPS can be written as

$$\begin{aligned} \mathrm{LPIPS}(Y,\hat{Y}) &= \sum_l \frac{1}{H_l W_l}\sum_{h,w} d_l(h,w), \\ d_l(h,w) &= \left\| w_l \odot \left(\phi_l(Y)_{hw} - \phi_l(\hat{Y})_{hw}\right)\right\|_2^2. \end{aligned} \tag{5}$$

where H_l and W_l are the spatial dimensions of the feature map at layer l, and $\odot$ denotes channel-wise weighting. Lower LPIPS indicates that the restored image is perceptually closer to the reference image.

These metrics capture different aspects of restoration quality. PSNR emphasizes pixel-level fidelity, SSIM emphasizes structural similarity, SAM emphasizes spectral consistency, ERGAS summarizes relative band-wise reconstruction error, and LPIPS emphasizes perceptual similarity in the visible bands. Since no single metric fully describes restoration quality, the framework reports multiple metrics and pairs them with qualitative visual outputs.

E. Visualization, Logging, and Extensibility

The framework generates visual outputs alongside numerical metrics. These outputs may include input-prediction-target panels, model comparison panels, zoomed crops, and difference maps. Visual inspection is necessary because restoration artifacts, such as blurring, boundary errors, spectral distortion, or unrealistic textures, may not be fully captured by aggregate metric values.

Each experiment also produces configuration records containing the selected dataset, model settings, preprocessing choices, metric options, and output paths. The metric files, visual outputs, and configuration records are stored together. This makes each run easier to inspect, reproduce, and compare with later experiments.

The framework is designed to be extended through modular loaders, model adapters, metric functions, and visualization modules. New datasets can be added by defining how their samples are loaded and mapped into the evaluation workflow.

New models can be added by defining how inputs are passed to the model and how outputs are returned. This makes the pipeline suitable for gradual expansion beyond the current AllClear and PROBA-V SR implementation.

IV. Experimental Setup

This section describes the concrete experimental instantiation of the proposed pipeline. The goal is not to establish a new state-of-the-art result, but to verify that the framework can organize restoration outputs, compute or collect metrics, generate visual summaries, and store experiment records in a reproducible form.

The super-resolution part of the experiment uses nine uploaded PROBA-V SR examples produced by Residual Attention Multi-image Super-Resolution (RAMS). For each example, the pipeline stores the low-resolution input, the restored output, the high-resolution reference, and the corresponding metric values.

The cloud-removal part of the experiment uses selected uploaded AllClear result panels. The available panels compare MultiTemporal, noSAR, and U-Net with Temporal Attention Encoder (UTAE) outputs for selected regions of interest (ROIs). These outputs are used to test whether the framework can organize model-level comparisons, report available metrics, and produce qualitative panels in a consistent format.

For each task, the pipeline follows the same high-level procedure. It loads the available inputs, predictions, references, and metric files; applies the required task-specific formatting; aligns outputs with their corresponding scene or ROI identifiers; and stores the resulting quantitative and visual outputs. For PROBA-V SR, the pipeline reports scene-level metrics and aggregates them into mean and standard deviation values over the nine available examples. For AllClear, the pipeline reports the available panel-level metrics for each ROI-model pair.

The pipeline produces quantitative tables, summary statistics, and qualitative visual panels. For PROBA-V SR, the generated outputs include a compact table over the nine RAMS examples and a visual grid of the selected scenes. For AllClear, the generated outputs include an ROI-by-model metric table and qualitative panels comparing the available cloud-removal outputs. This allows each run to be inspected after execution and makes later extension to more scenes, models, and datasets straightforward.

V. Results and Discussion

This section reports the quantitative and qualitative outputs generated by the proposed benchmarking pipeline. The results are organized by task. The super-resolution results are reported first because the available PROBA-V SR outputs include scene-level metric files. The cloud-removal results are then reported using the available AllClear panel-level outputs.

A. PROBA-V SR Quantitative Results

Table I summarizes the nine available PROBA-V SR examples produced by RAMS. The reported metrics are PSNR and corrected PSNR (cPSNR), SAM, and ERGAS. Higher PSNR, SSIM, and cPSNR indicate better reconstruction quality, while lower SAM and ERGAS indicate lower spectral or relative reconstruction error.

TABLE I

Quantitative results on the nine available PROBA-V SR examples produced by Residual Attention Multi-image Super-Resolution (RAMS). Higher PSNR, SSIM, and cPSNR indicate better reconstruction quality, while lower SAM and ERGAS indicate lower error.

Scene ID	PSNR (dB) ↑	SSIM ↑	cPSNR (dB) ↑	SAM ↓	ERGAS ↓
imgset0000	42.3407	0.9811	45.5316	0.0693	5.0514
imgset0001	45.1622	0.9765	38.3062	0.1380	4.9434
imgset0002	44.5694	0.9791	40.6762	0.0747	2.8134
imgset0003	44.7266	0.9740	39.0688	0.0794	2.7213
imgset0004	47.7315	0.9854	45.4047	0.0728	5.4121
imgset0005	39.1411	0.9488	40.1394	0.0889	3.7014
imgset0006	44.8080	0.9726	38.5797	0.0803	2.7324
imgset0007	37.8149	0.9201	36.2513	0.0934	3.7022
imgset0008	42.1001	0.9911	43.9466	0.0560	2.8484
Mean (μ)	43.1549	0.9698	40.8783	0.0836	3.7696
Std Dev (σ)	3.1341	0.0221	3.3303	0.0231	1.0992

Across the nine examples, the mean PSNR is 43.1549 dB, the mean SSIM is 0.9698, and the mean cPSNR is 40.8783 dB. These values indicate strong reconstruction quality for the selected outputs. However, the standard deviations show that performance varies across scenes. In particular, the variation in cPSNR, SAM, and ERGAS suggests that some scenes remain more difficult than others, even when processed through the same evaluation workflow.

These results support the framework-level claim of the paper. The pipeline is able to organize scene-level super-resolution outputs, compute or collect multiple reconstruction metrics, aggregate them into summary statistics, and connect the numerical results with visual artifacts. The current results should be interpreted as a sample-level validation of the pipeline rather than as a full PROBA-V SR leaderboard comparison.

B. AllClear Quantitative Results

Table II summarizes the available AllClear panel-level metrics for selected ROI and model pairs. The compared models are MultiTemporal, noSAR, and UTAE. The reported metrics are PSNR, SSIM, SAM, LPIPS, as well as mean absolute error (MAE), and root mean squared error (RMSE). The results show that model behavior varies by ROI and by metric. For roi43484_0044, noSAR obtains the highest PSNR and the lowest SAM, MAE, RMSE, and LPIPS, while MultiTemporal obtains the highest SSIM. For roi6519_0066, UTAE obtains the highest PSNR and the lowest SAM, MAE, RMSE, and LPIPS, while MultiTemporal obtains the highest SSIM. This shows why a benchmarking pipeline should report multiple metrics rather than relying on a single score.

These results should be interpreted as panel-level evidence from the current experiment outputs. They demonstrate that the framework can organize ROI-level model comparisons and present them consistently, but they do not establish a full

TABLE II
QUANTITATIVE RESULTS ON AVAILABLE ALLCLEAR PANEL-LEVEL EXAMPLES. HIGHER PSNR AND SSIM INDICATE BETTER RECONSTRUCTION QUALITY, WHILE LOWER SAM, MAE, RMSE, AND LPIPS INDICATE LOWER ERROR.

ROI ID	Model	PSNR ↑	SSIM ↑	SAM ↓	MAE ↓	RMSE ↓	LPIPS ↓
roi43484_0044	MultiTemporal	11.65	0.757	58.46	0.1285	0.2615	0.703
roi43484_0044	noSAR	22.74	0.729	30.73	0.0602	0.0730	0.514
roi43484_0044	UTAE	14.29	0.745	52.45	0.1050	0.1929	0.625
roi6519_0066	MultiTemporal	15.12	0.440	63.17	0.1208	0.1753	0.276
roi6519_0066	noSAR	15.95	0.369	52.09	0.1367	0.1594	0.292
roi6519_0066	UTAE	18.02	0.412	49.58	0.1065	0.1256	0.225
Mean (μ)		16.2950	0.5753	51.0800	0.1096	0.1646	0.4392
Std Dev (σ)		3.7844	0.1860	11.1346	0.0272	0.0636	0.2019

AllClear benchmark ranking. Figure 1 complements the table by aggregating the available AllClear results at the model level. This view makes cross-metric trade-offs easier to see: the noSAR variant achieves the lowest MAE, RMSE, SAM, and LPIPS together with the highest PSNR, while MultiTemporal attains the highest SSIM. UTAE remains between these two behaviors on most metrics, which shows that model ranking depends on which restoration criterion is emphasized.

C. Qualitative Results

Qualitative inspection is important because restoration metrics do not always capture all visible artifacts. Blurring, boundary errors, spectral distortion, texture artifacts, and residual cloud effects may be easier to identify visually than from aggregate metric values alone. The framework therefore stores qualitative outputs together with quantitative summaries.

The qualitative outputs provide visual context for the metric tables. For AllClear, the generated panels compare MultiTemporal, noSAR, and UTAE on the same ROIs. This layout makes model differences easier to inspect because each row corresponds to the same scene and each column corresponds to a different model output. For PROBA-V SR, Figure 2 groups four representative RAMS examples in a single compact layout so that all selected scenes remain together. Each image shows the low-resolution input, the RAMS prediction, and the high-resolution reference from left to right.

D. Discussion

The results support two main observations. First, the proposed pipeline can organize outputs from two different restoration settings under a shared reporting structure. It produces metric tables, summary statistics, qualitative figures, and configuration-linked artifacts. This supports the main methodological claim of the paper: the contribution is a reusable benchmarking workflow rather than a new restoration architecture.

Second, the results show that model interpretation depends on both scene content and metric choice. In the AllClear examples, the model with the best SSIM is not always the model with the best PSNR, SAM, MAE, RMSE, or LPIPS. This supports the use of multiple complementary metrics and qualitative outputs in restoration evaluation.

The current results also define the limits of the paper. The PROBA-V SR analysis uses nine RAMS examples, and the AllClear analysis uses selected panel-level outputs. Therefore, the paper does not claim to be a full benchmark or complete model ranking. The evidence is sufficient to demonstrate framework usability and reproducibility, while broader empirical validation remains future work.

VI. CONCLUSION AND FUTURE WORK

This paper presented a compact evaluation pipeline for satellite image restoration experiments. The pipeline brings together dataset loading, preprocessing, model execution, metric computation, visualization, and result logging in a single workflow. The current implementation focuses on two representative restoration settings: cloud removal with AllClear and super-resolution with PROBA-V SR.

The main contribution is not a new restoration model, but a reusable experimental structure that makes restoration outputs easier to evaluate, compare, and report. By combining standard image-quality metrics with remote-sensing-oriented measures and visual outputs, the pipeline provides a practical basis for clearer experimental analysis. The reported results should be interpreted as an initial demonstration of the pipeline rather than a complete benchmark, especially because the PROBA-V SR evaluation is limited to the available examples.

Future work will extend the pipeline with broader model coverage, larger evaluation sets, and more complete benchmark summaries for both supported tasks. Further improvements may include richer visualization tools, more automated experiment tracking, and stronger aggregation of metric outputs across datasets, scenes, and restoration settings. These extensions would make the pipeline more useful as a reusable evaluation tool for satellite image restoration research.

DECLARATIONS

The authors declare that no funding was received for the conduct of this research.

AUTHOR CONTRIBUTIONS

RH: Contributed to pipeline design and architecture, dataset integration, metric integration, the visualization pipeline, and literature review. NA: Contributed to dataset preparation and integration, preprocessing pipeline development, evaluation implementation, and metric integration. PZ: Contributed to data analysis and validation, project website development, repository organization, and literature review. HS: Contributed to visualization and experiments, model comparison outputs, website development, and literature review.

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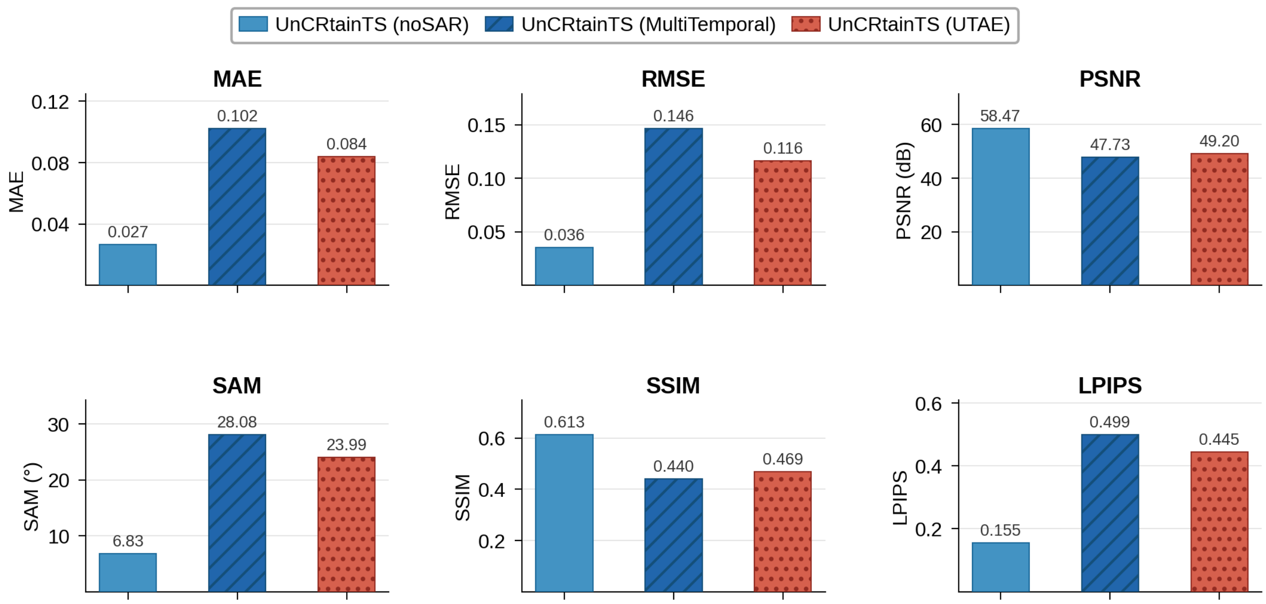

Fig. 1. Aggregated AllClear model comparison across six evaluation metrics for the available examples. Lower values are better for MAE, RMSE, SAM, and LPIPS, while higher values are better for PSNR and SSIM. The chart summarizes the same model families reported in Table II and shows that the noSAR configuration provides the strongest overall error profile, whereas MultiTemporal attains the highest SSIM.

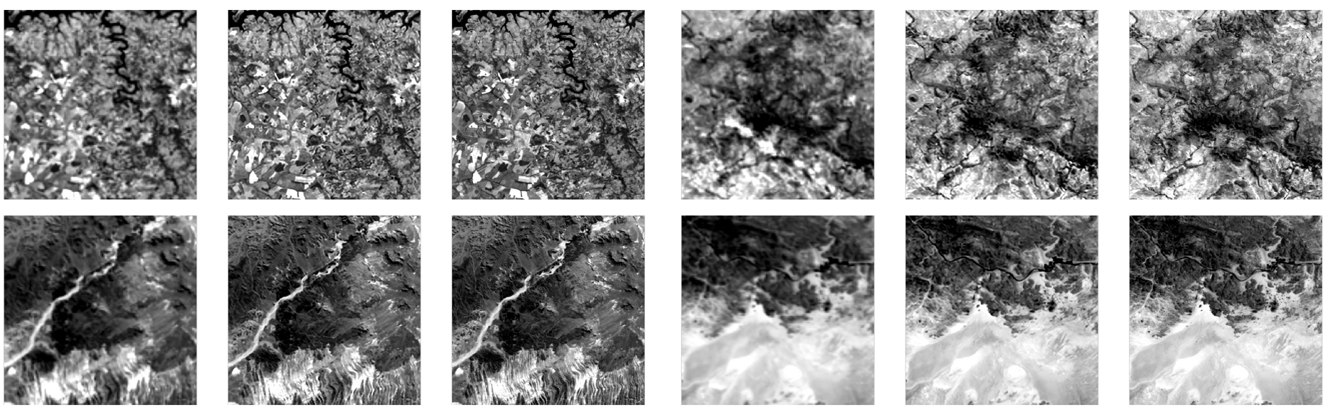

Fig. 2. Qualitative PROBA-V SR results for four representative RAMS scenes. In each composite, the three panels show the low-resolution input, the RAMS reconstruction, and the high-resolution reference. Top left: `imgset0001`. Top right: `imgset0002`. Bottom left: `imgset0006`. Bottom right: `imgset0007`.

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